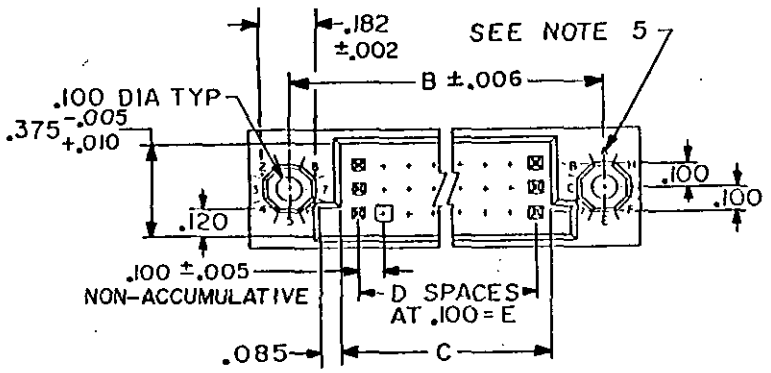
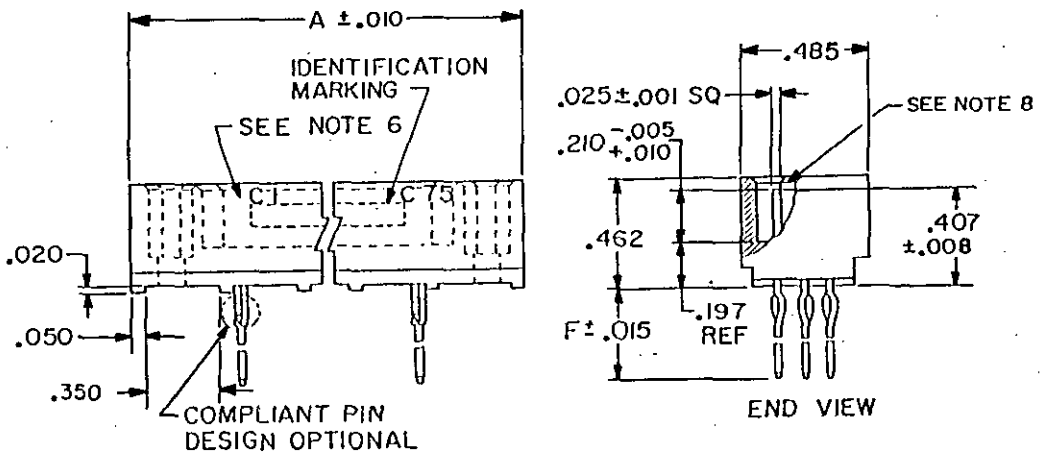
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 3-ROW PIN ASSEMBLIES, 75 THROUGH 405 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.





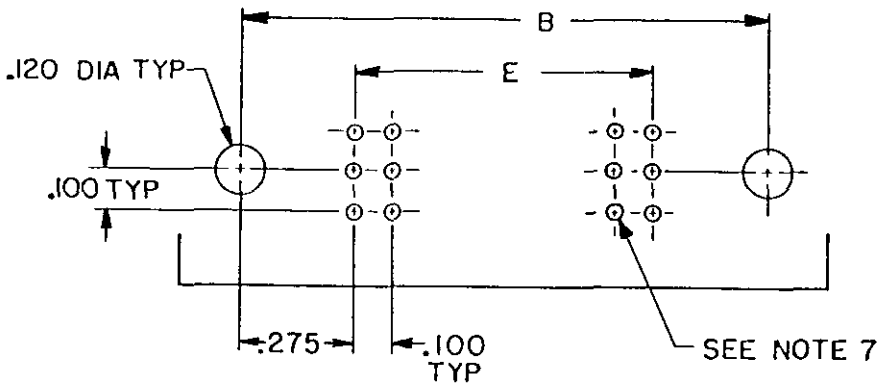
F ±.015		Type	Type of termination
inches	mm		
.733	18.62	I	Compliant pin
.533	13.54	II	"
.328	8.33	III	"
.250	6.35	IV	"
.177	4.50	V	"
.190	4.83	VI	Solder post 1/
.165	4.19	VII	"
.140	3.56	VIII	"
.120	3.05	IX	"

1/ Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 75 through 225 contacts.

AMSC N/A
FSC 5935

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.050	1.27
.085	2.16
.100	2.54
.120	3.05
.182	4.62
.190	4.83
.197	5.00
.210	5.33
.250	6.35
.275	6.98
.350	8.89
.375	9.53
.407	10.34
.462	11.73
.485	12.32
.533	13.54
.733	18.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/177.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and lines every ten cavities stamped this surface.
7. Holes for solder type shall be .042-.050 (1.067-1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ±.0010 (1.15 ±0.0254 mm) drilled hole (1.15 mm drill) .001-.003 (0.025-0.076 mm) thick copper on wall, .0003 (0.0076 mm) minimum tin lead plating, finished hole diameter .037-.043 (0.94-1.09 mm) after plating, .036-.043 (0.91-1.09 mm) after reflow, copper hardness (Knoop) 150 maximum.
8. Chamfer shall be .015 minimum at a 45° angle ±5 degrees.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 75 through 225 contacts - Continued.

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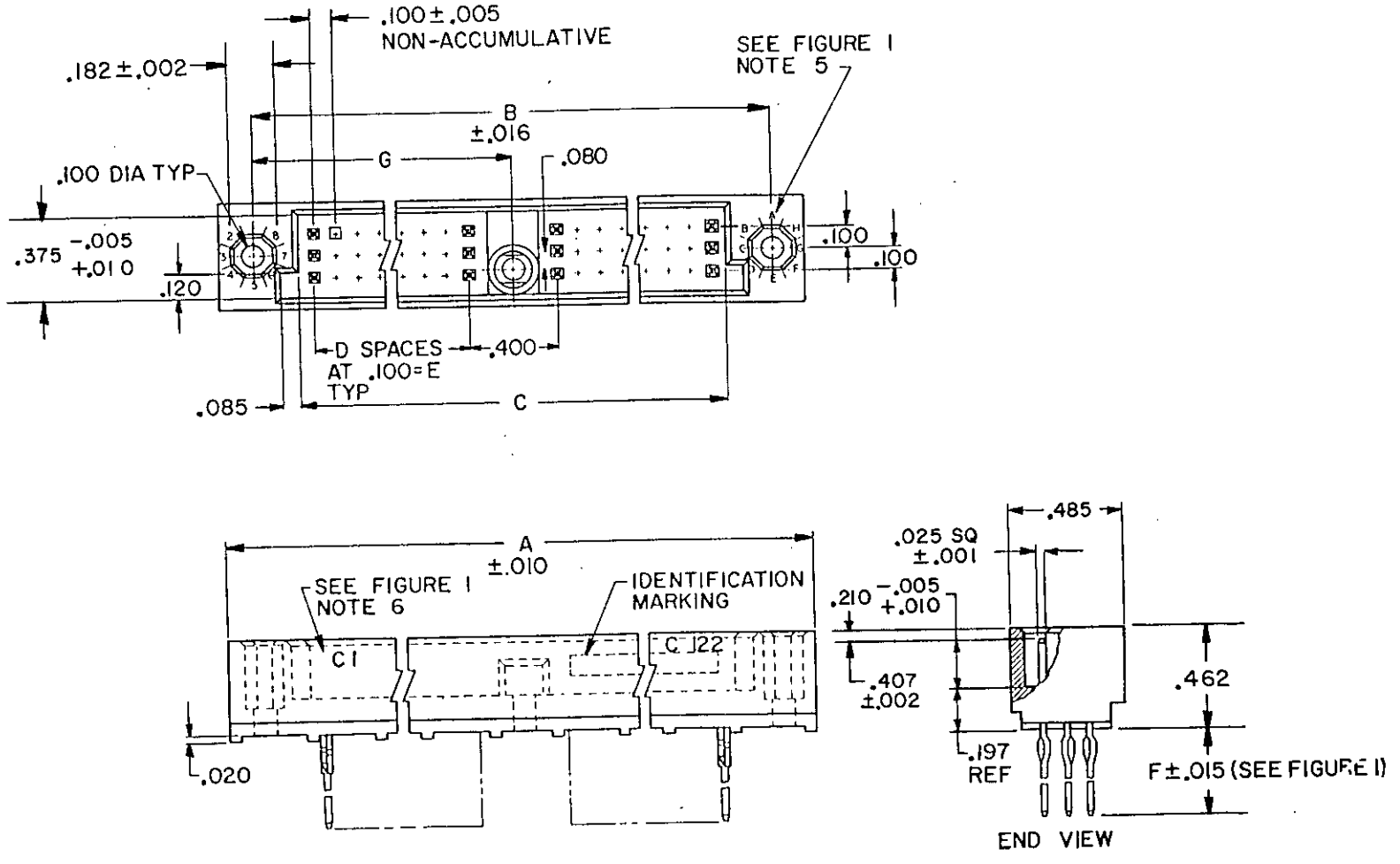
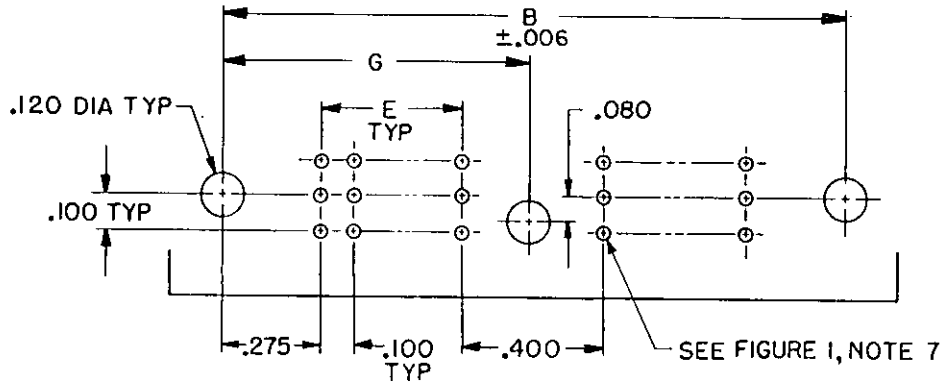


FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 240 through 366 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.080	2.03
.085	2.16
.100	2.54
.120	3.05
.152	3.86
.182	4.62
.197	5.00
.210	5.33
.275	6.98
.310	7.87
.350	8.89
.375	9.53
.400	10.16
.407	10.34
.462	11.73
.485	12.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 240 through 366 contacts - Continued.

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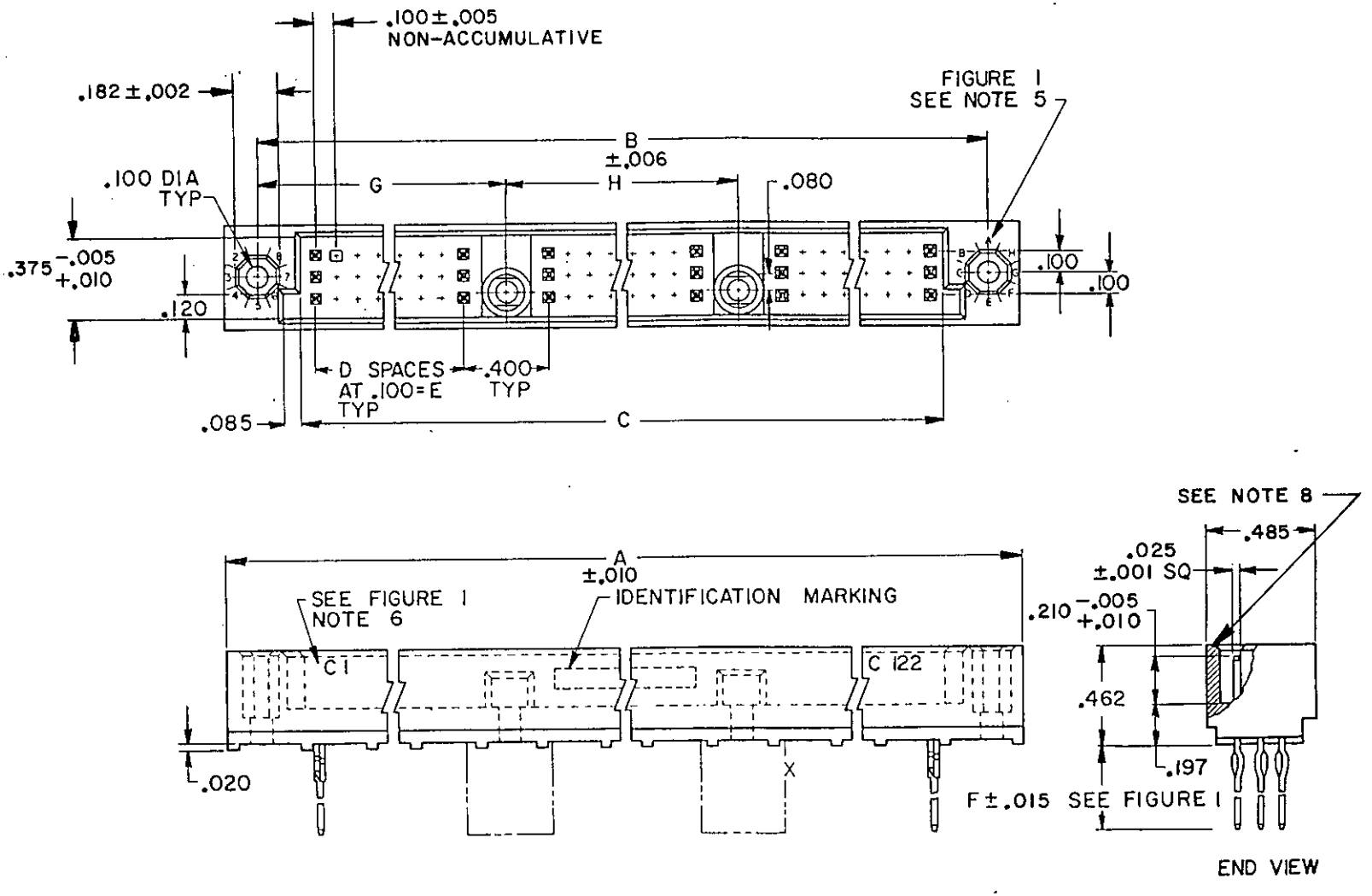
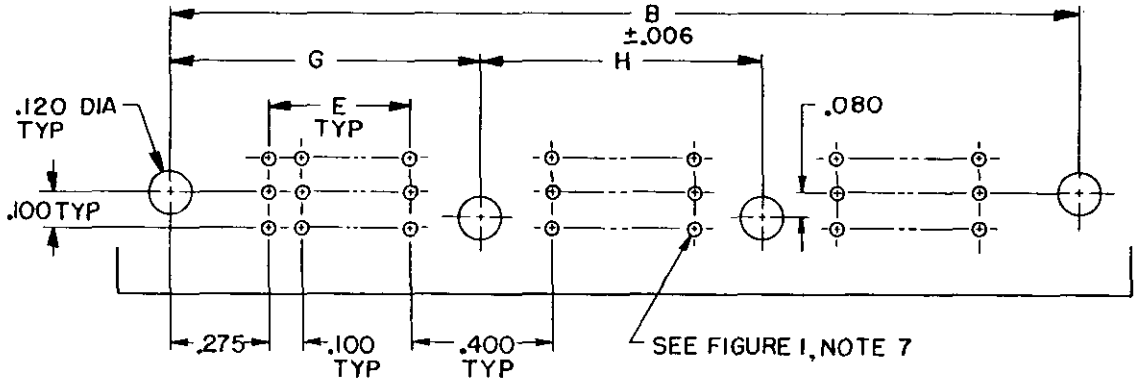


FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 387 through 405 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.080	2.03
.085	2.16
.100	2.54
.120	3.05
.152	3.86
.182	4.62
.197	5.00
.210	5.33
.275	6.98
.310	7.87
.375	9.53
.400	10.16
.462	11.73
.485	12.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With two guides, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 387 through 405 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 03 04 05 06 07 08 09	75	1	I II III IV V VI VII VIII IX	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
10 11 12 13 14 15 16 17 18	90	1	I II III IV V VI VII VIII IX	3.750 (95.25)	3.450 (87.63)	3.050 (77.47)	29	2.900 (73.66)			1-30
19 20 21 22 23 24 25 26 27	96	1	I II III IV V VI VII VIII IX	3.950 (100.03)	3.650 (92.71)	3.250 (82.55)	31	3.100 (78.74)			1-32
28 29 30 31 32 33 34 35 36	105	1	I II III IV V VI VII VIII IX	4.250 (107.95)	3.950 (100.33)	3.550 (90.17)	34	3.400 (86.36)			1-35
37 38 39 40 41 42 43 44 45	120	1	I II III IV V VI VII VIII IX	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
46 47 48 49 50 51 52 53 54	135	1	I II III IV V VI VII VIII IX	5.250 (133.35)	4.950 (125.73)	4.550 (115.57)	44	4.400 (111.76)			1-45
55 56 57 58 59 60 61 62 63	150	1	I II III IV V VI VII VIII IX	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
64 65 66 67 68 69 70 71 72	165	1	I II III IV V VI VII VIII IX	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)			1-55
73 74 75 76 77 78 79 80 81	180	1	I II III IV V VI VII VIII IX	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
82 83 84 85 86 87 88 89 90	195		I II III IV V VI VII VIII IX	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)			1-65

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
91 92 93 94 95 96 97 98 99	201	1	I II III IV V VI VII VIII IX	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
100 101 102 103 104 105 106 107 108	210	1	I II III IV V VI VII VIII IX	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70
109 110 111 112 113 114 115 116 117	225	1	I II III IV V VI VII VIII IX	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)			1-75
118 119 120 121 122 123 124 125 126	240	2	I II III IV V VI VII VIII IX	9.050 (229.87)	8.750 (222.25)	8.350 (212.09)	39	3.900 (99.06)	4.375 (111.13)		1-80
127 128 129 130 131 132 133 134 135	258	2	I II III IV V VI VII VIII IX	9.650 (245.11)	9.350 (237.49)	8.950 (227.33)	42	4.200 (106.68)	4.675 (118.75)		1-86

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
136	276	2	I	10.250	9.950	9.550	45	4.500	4.975		1-92
137			II	(260.35)	(252.73)	(242.57)		(114.30)	(126.37)		
138			III								
139			IV								
140			V								
141			VI								
142			VII								
143			VIII								
144			IX								
145	294	2	I	10.850	10.550	10.150	48	4.800	5.275		1-98
146			II	(275.59)	(267.97)	(257.81)		(121.92)	(133.99)		
147			III								
148			IV								
149			V								
150			VI								
151			VII								
152			VIII								
153			IX								
154	312	2	I	11.450	11.150	10.750	51	5.100	5.575		1-104
155			II	(290.83)	(283.21)	(273.05)		(129.54)	(141.61)		
156			III								
157			IV								
158			V								
159			VI								
160			VII								
161			VIII								
162			IX								
163	330	2	I	12.050	11.750	11.350	54	5.400	5.875		1-110
164			II	(306.07)	(298.45)	(288.29)		(137.16)	(149.23)		
165			III								
166			IV								
167			V								
168			VI								
169			VII								
170			VIII								
171			IX								
172	348	2	I	12.650	12.350	11.950	57	5.700	6.175		1-116
173			II	(321.31)	(313.69)	(303.53)		(144.78)	(156.85)		
174			III								
175			IV								
176			V								
177			VI								
178			VII								
179			VIII								
180			IX								

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
181	366	2	I	13.250	12.950	12.550	60	6.000	6.475		1-122
182			II	(336.55)	(328.93)	(318.77)		(152.40)	(164.47)		
183			III								
184			IV								
185			V								
186			VI								
187			VII								
188			VIII								
189			IX								
190	387	3	I	14.250	13.950	13.550	42	4.200	4.675	4.600	1-129
191			II	(361.95)	(354.33)	(344.17)		(106.68)	(118.75)	(116.84)	
192			III								
193			IV								
194			V								
195			VI								
196			VII								
197			VIII								
198			IX								
199	405	3	I	14.850	14.550	14.150	44	4.400	4.875	4.800	1-135
200			II	(377.19)	(369.57)	(359.41)		(111.76)	(123.83)	(121.92)	
201			III								
202			IV								
203			V								
204			VI								
205			VII								
206			VIII								
207			IX								

TABLE II. Contact plating requirements.

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area, .170 inch minimum length, shall be gold over nickel in accordance with MIL-C-55302. Compliant pin area (contact type I, .748 inch minimum length; type II, .548 minimum length; type III, .343 inch minimum length; type IV, .265 inch minimum length; type V, .192 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302. Solder post area (contact type VI, .205 minimum length; type VII, .190 inch minimum length; type VIII, .165 inch minimum length; type IX, .135 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (86/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290, and shall meet the requirements of MIL-P-45209.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, 3, and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-C-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Compliant section shall meet the requirements of MIL-STD-2166, type III. For use with .093 minimum thick printed wiring boards. Sequence of tests and number of samples shall be in accordance with MIL-STD-2166, table III.

Wire wrappost: See MIL-STD-1130.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .094; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number: M55302/176 (and dash number from table I and type from table II).

Example: M55302/176-01A.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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